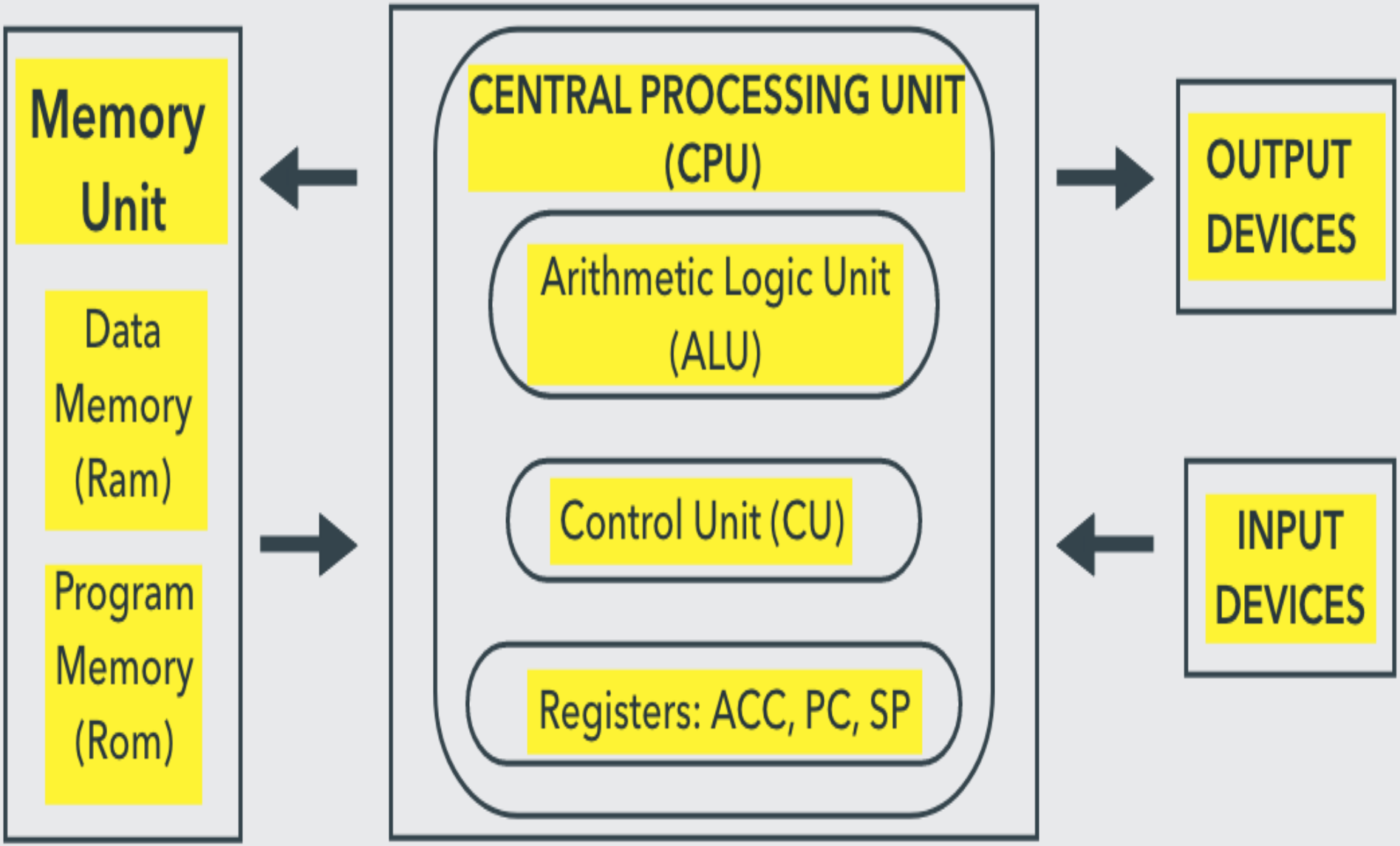
Introduction

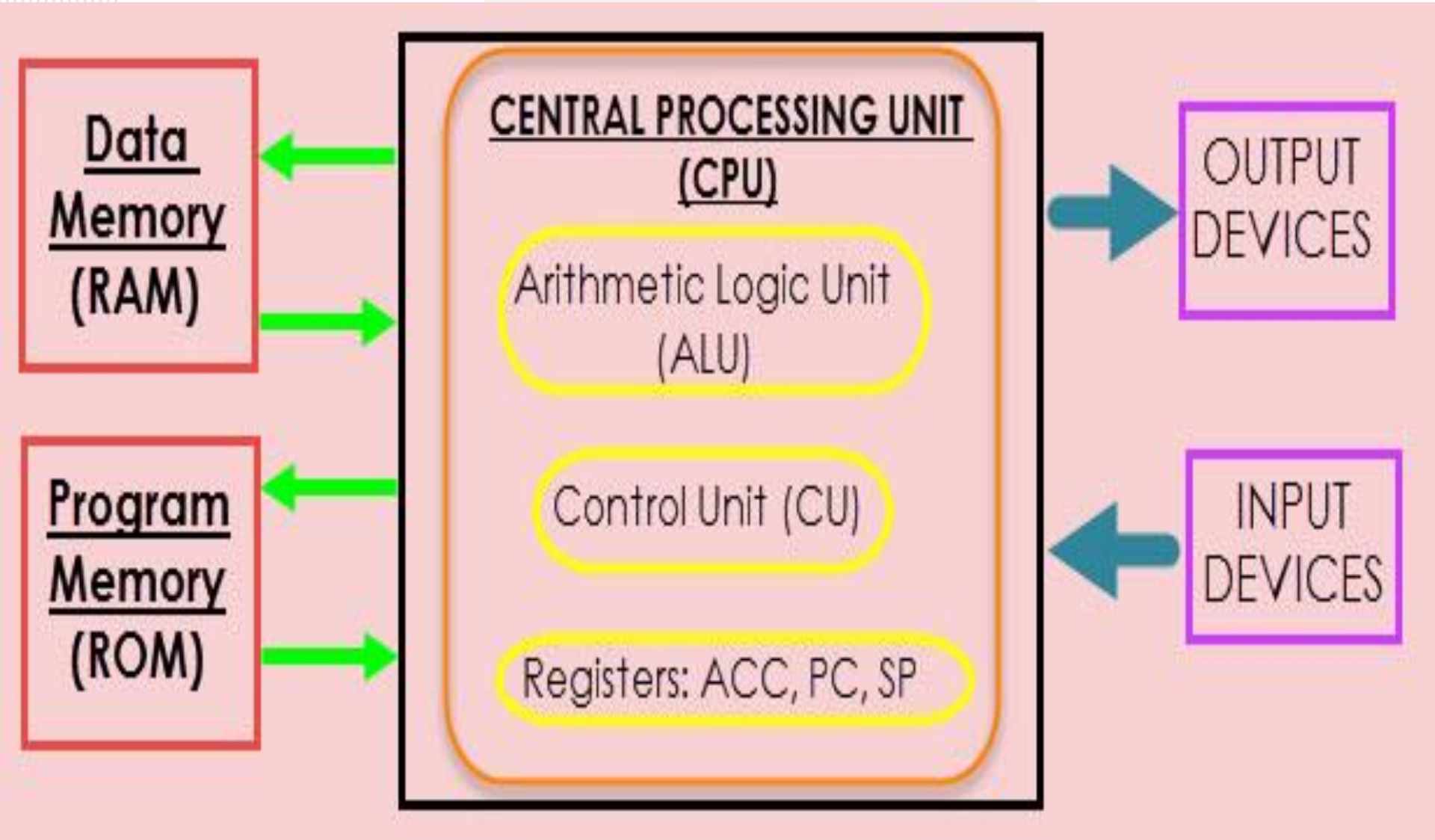
- CPU Design Architectures (Schools of Thought)
- Based On Memory Organization
  - Von Neuman/Princeton Architecture
  - Harvard Architecture
- Based On Instruction Set Architecture
  - CISC Architecture
  - RISC Architecture

Von Neumann Architecture



Electronics Hub

Harvard Architecture





CISC

- CISC – Complex Instruction Set Computing.
- Large No. of instructions.
- Software burden is less
- IDU size is more .
- Power Consumption is more.
- Less no. of internal CPU registers
- processing is memory intensive.
- Less Efficient & Slow
- Instructions of Varying Sizes, & Varying Timings for Fetch & Execution clock cyles
- Many Addressing Modes
- Partial Orthogonality
- Pipelining Optional
- Microcode/programmed CU

RISC

- RISC – Reduced Instruction Set Computing.
- Relatively Less No. Of instructions .
- Software burden is more
- IDU size is less comparatively.
- Power consumption is less comparatively.
- More no. of internal CPU registers
- processing is register intensive.
- More Efficient & Fast
- Instructions of Fixed Sizes, Same Fetch time & Mostly Same Execution cycles.
- Few Addressing Modes
- Full Orthogonality
- Pipelining MUST be Used
- Hardwired CU